

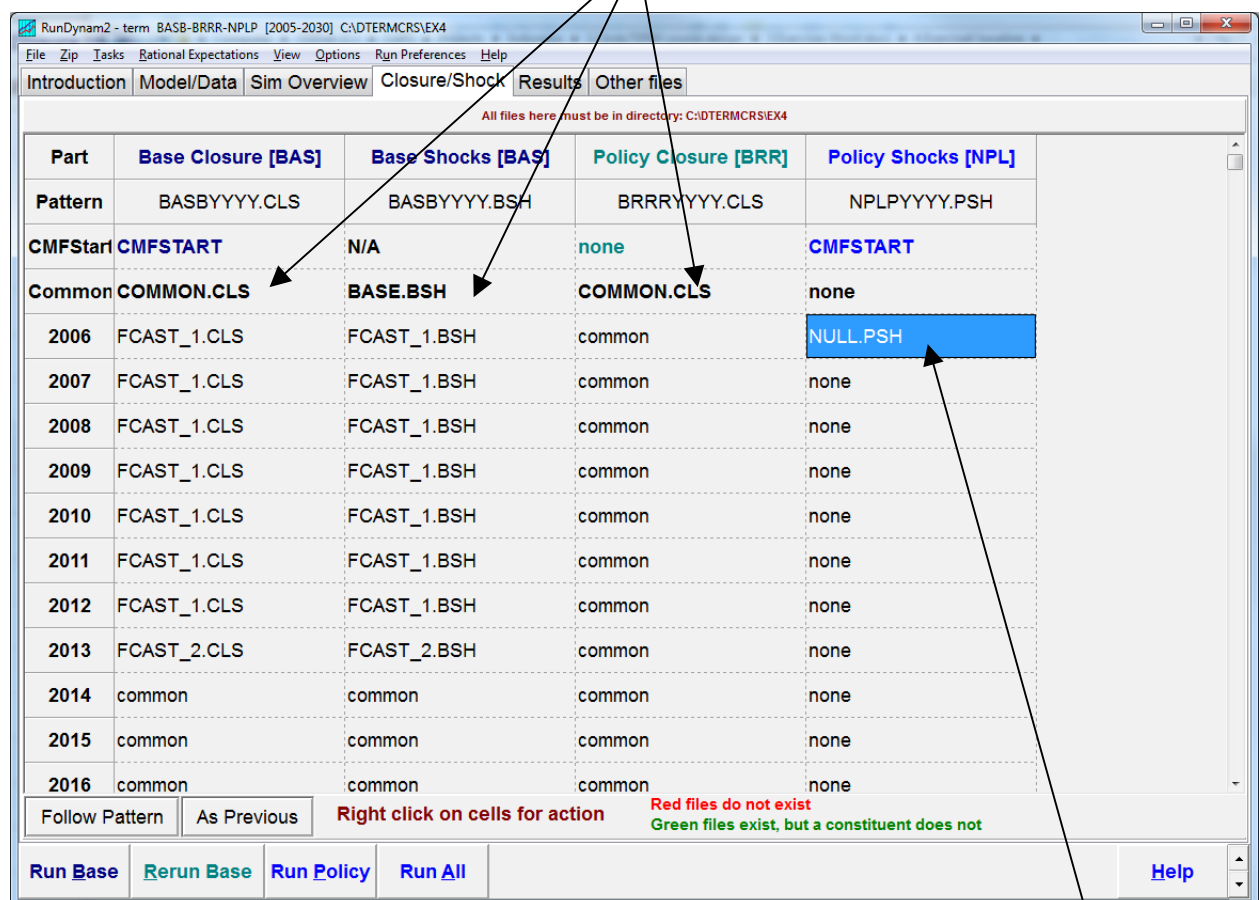
Exercise 5:

Moratorium on land use in the OilPalm industry

In this exercise you will establish a policy shock and run a dynamic simulation. The simulation reports the results of a moratorium place on land cleared for the cultivation of OilPalm. You have already done the base and base-rerun in your previous lab session. All you need to do is set up policy shock file (.PSH) for the years 2014-2030 that introduces a shock to the use of land in the OilPalm sector.*

1. Getting started

Open RunDynam. Your previous simulation (the baseline exercise) should still be open. This OilPalm simulation uses the results from the baseline exercise as its base case.



2. The Simulation

In the baseline simulation (EX4), no policy shocks were specified. We used the file NULL.PSH in the "Policy Shocks" column. In this exercise, you need to introduce a policy shock (*.PSH) file into the simulation.

To create the OilPalm policy shock file, right click on NULL.PSH and select **Edit**. In the NULL.PSH file you will find one shock statement, which is:

```
aShock delunity=0;
```

Delete this statement. Now type the following statement:

```
tshock xln("OilPalm",DST) = uniform 0;
```

Save this file as **OILPALM.PSH**.

Notice that the shock statement is a “t” shock. To explain the shock, a little background is necessary.

Recursive-dynamic models are multi-period CGE models in which results are computed one-period-at-a-time. RunDynam is a program, sold separately from the rest of GEMPACK, designed to manage simulations with these recursive-dynamic models.

The RunDynam user does not personally create CMF files for each year (as is usual for standard GEMPACK models). Rather RunDynam creates the many CMF files which are needed, using information from the user about closure and shocks in each year.

A recursive-dynamic simulation yields raw results which are a sequence of annual percentage (or maybe ordinary) changes. Often there are two such sequences or simulations:

- a 'base', 'control' or 'business-as-usual' run which is a forecast driven by many expected exogenous changes stored in BSH (base shock) files created by the user. The shock values are usually expressed as changes relative to the value in the preceding year.
- a 'policy' or 'perturbed' scenario which is the same as the base except that the exogenous changes include some extra shocks of policy interest, such as a tariff change proposed for 2014. The extra shocks are stored in PSH (policy shock) files created by the user.

Each simulation produces a sequence of year-on-year changes in all model variables. Results (the effect of the proposed tariff change) are normally reported as cumulative differences (for any particular year) between the two scenarios, eg, as the percent difference between employment in 2019 with the tariff change relative to employment in 2019 without the change. This entails a great deal of tedious calculation, which RunDynam or ViewSOL does behind the scenes.

Of course the tariff may have changed in both scenarios. In that case the cumulative difference shows the effect of additional tariff changes implemented for the policy scenario. It is natural and convenient to think of the policy shock in the same way -- as the extra component of tariff change, over and above any tariff change included in the base scenario. Indeed usually the RunDynam user specifies policy shocks in a PSH file with statements like:

```
ashock t0imp = uniform 3;
```

meaning that t0imp experiences a year-on-year increase of 3%, on top of any change in the base scenario. Suppose that in the base scenario for that year t0imp increased uniformly by 2%. Then, in the CMF file that RunDynam would construct for that year's policy simulation we would find the statements:

```
shock t0imp = uniform 2;
```

... many other statements

```
ashock t0imp = uniform 3;
```

GEMPACK compounds the two shocks, and will actually apply a year-on-year increase of 5.06% ($1.02 \times 1.03 = 1.0506$). Usually the result is close to simple addition of the shocks ($5 = 2 + 3$). The **ashock** statement means "additional shock".

The point to remember is that policy shock statements in PSH files are included in the CMF after a long list of shocks inherited from the base simulation, which potentially includes shocks to every exogenous variable. Furthermore, you are not allowed in a GEMPACK CMF file to shock the same variable twice using the same type of shock statement. Hence it would be illegal to include in your PSH file a statement like:

```
shock t0imp = uniform 6;
```

to cause t0imp to grow by 6% in the policy scenario, regardless of any change in the base scenario. If you did, RunDynam would construct a CMF file including the statements:

```
shock t0imp = uniform 2; ! shock from base
```

... many other statements

`shock t0imp = uniform 6; ! second "shock" not allowed`

causing an error. To achieve that effect (grow 6% in policy whatever happened in base), the PSH file should include the statement:

`tshock t0imp = uniform 6;`

so the CMF file includes the statements:

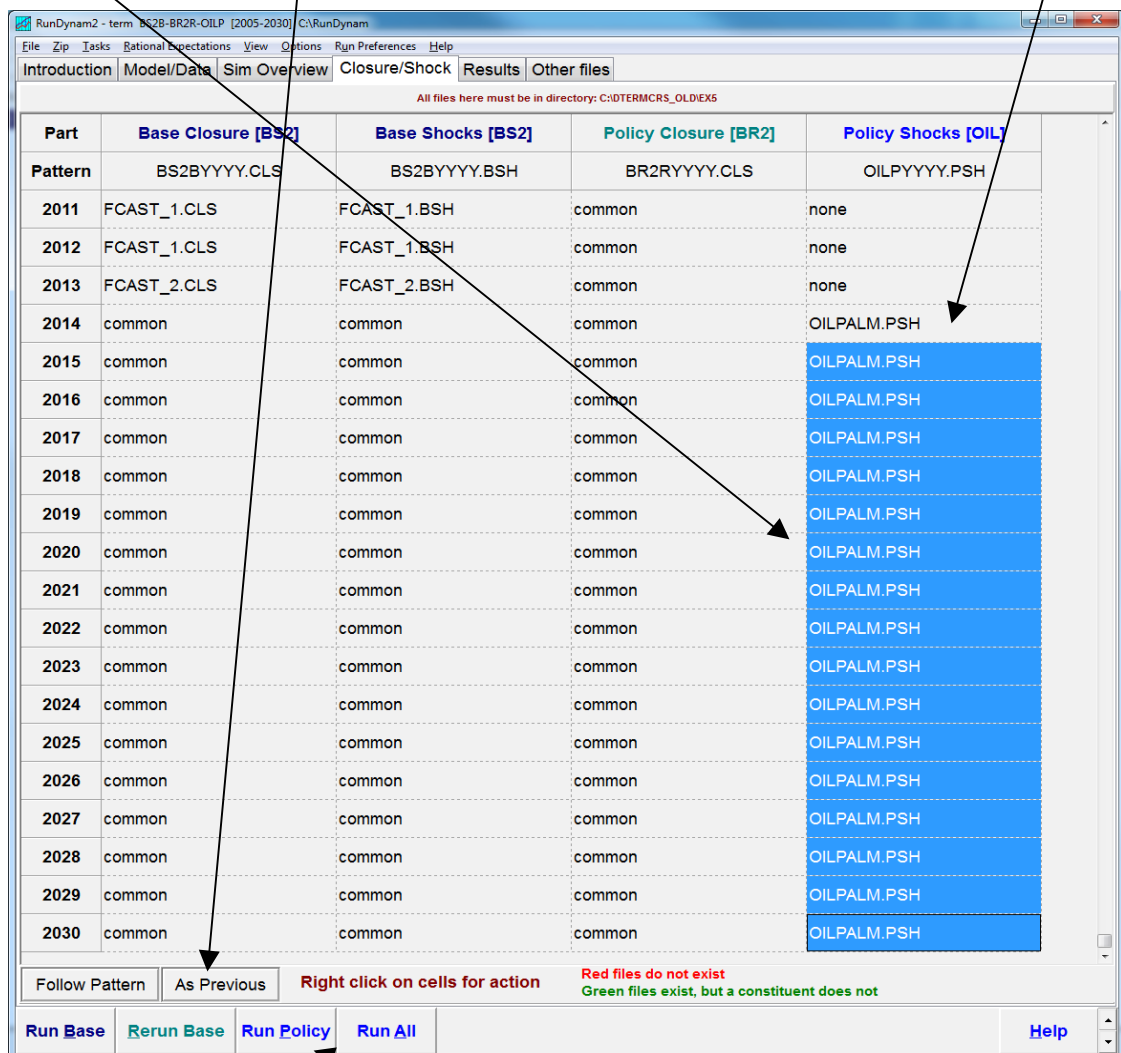
`shock t0imp = uniform 2;`

... many other statements

`tshock t0imp = uniform 6; ! ignore previous shock`

In this case the base shock (2%) is simply ignored. The **tshock** statement means "target shock".

Now go to the "Policy Shocks" column and select OILPALM.PSH for all the years starting at 2014 and ending with 2030. An easy way to specify the policy shocks is to select OILPALM.PSH for **2014**. Then **highlight** the years 2015-2030 by right clicking on the mouse and selecting all the years from 2015 to 2030. Now click on the "**As Previous**" button in the left bottom corner of Rundynam. Your Closure/Shock page should look like this:



Check for errors in your policy closure and shocks by selecting **Tasks | Check all Policy Closure and Shocks**.¹ To run the policy simulation select **Run Policy**.²

¹ If you have errors you can look at OILPALM.PSH in your "extra" folder to see a complete shock file.

3. Interpretation

Go to the Results page. To examine the difference between policy and basecase paths, you need to view results for all/several years (the top half of the screen). For “Which Run”, select Differences - Policy v Rerun, and for “Which Years” select All. You may only view cumulative results for the difference between the policy and rerun (cumulative will have been automatically selected when you chose to view Differences). Select Graph/ViewSol.

3.1 You have shocked the quantity of land used in the OilPalm sector over all regions by zero per cent. What does this mean?

3.2 Examine national employment / wage deviations.

Create a graph that contains deviations for *realwage_io* and *AggEmploy*. You will need to use the Charter tool in ViewSOL. Find the results for *realwage_io* and *AggEmploy* in the NatMacro matrix in ViewSOL. Explain the results.

3.3 Examine deviations for national employment, capital, land and real GDP.

Create a graph that contains deviations for national employment., aggregate capital stock, aggregate land and real GDP.

Plot a graph for *AggEmploy* (NatMacro), *AggCapStock* (NatMacro), *RealGDP* (NatMacro) and *xlnd_id* (Macros).

What accounts for the deviation path in *RealGDP*?

² You may receive messages warning you about incorrect prerequisite files, especially if you did not run the base or base rerun simulations in the previous exercise. Provided that you unzipped the results to your working directory, you should be able to proceed.

3.4 Plot deviation results for investments, the price of capital and the price of investment.

Create a graph that plots deviations for *pcap_id* (Macros), *INV* (pNatXGDPEXP) and *RealInv* (NatMacro).

3.5 Examine the deviation paths for GDP, GNE, imports, exports and the terms of trade.

Create a graph that plots deviations for *RealGNE* and *RealGDP*.

Create a graph that plots deviations for *RealGDP* (NatMacro), *ImpVolUse* (NatMacro), *ExpVol* (NatMacro) and *pGNE_pGDP* (bote).

3.6 Examine the deviation paths for real GDP, real private and public consumption, and real investment

Create a graph that plots deviations for *RealHou*, *RealInv*, *RealGov* and *RealGDP*.

3.7 Examine the deviation paths for region-specific GDP

Create a graph that plots *xgdpexp*.

Why does the moratorium on land use cause some regions to benefit while others contract? Which three regions benefits the least and the most from this policy?

3.8 Examine the deviation paths for state employment and land use.

Create a graph that plots *xlab_io*.

What accounts for the deviation paths of *xlab_io*?

Create a graph for *xlnd_i*.

Which regions shows the largest decline in land use? Can you explain why.

3.9 Examine the deviations for industry output in West Sumatra.

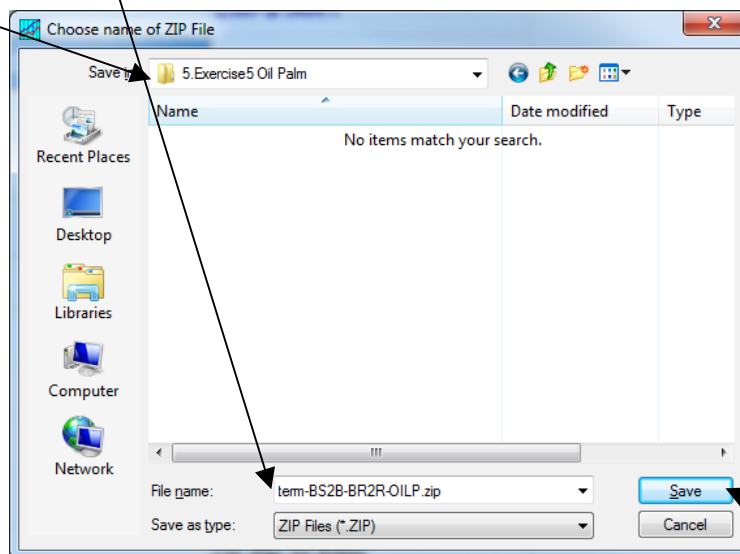
Create a graph that plots deviations for $xtotsho(*, WSumatra)$. Identify the three best and worst (apart from OilPalm) performing industries.

4. Saving simulation ingredients

You may want to save the simulation ingredients. Saving the simulation ingredients means that you will save:

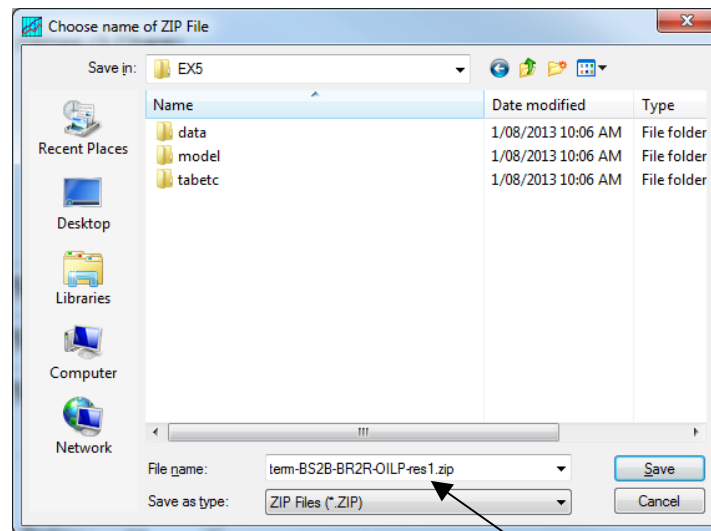
- The model (TERM.TAB, TERM.EXE) (Model/Data page)
- The database (AGGMOD.HAR, AGGSETS.HAR) (Model/Data page)
- Simulation names and *.ds1, solution method (Sim Overview page)
- The files specified in the Closure/Shock page. These files include all the *.CLS, *.BSH and *.PSH files.

Click on **Zip | Save ingredients as Zip archive**. Then click on **No**. You now have to save the *.zip file in a directory and choose a name for the zip file.



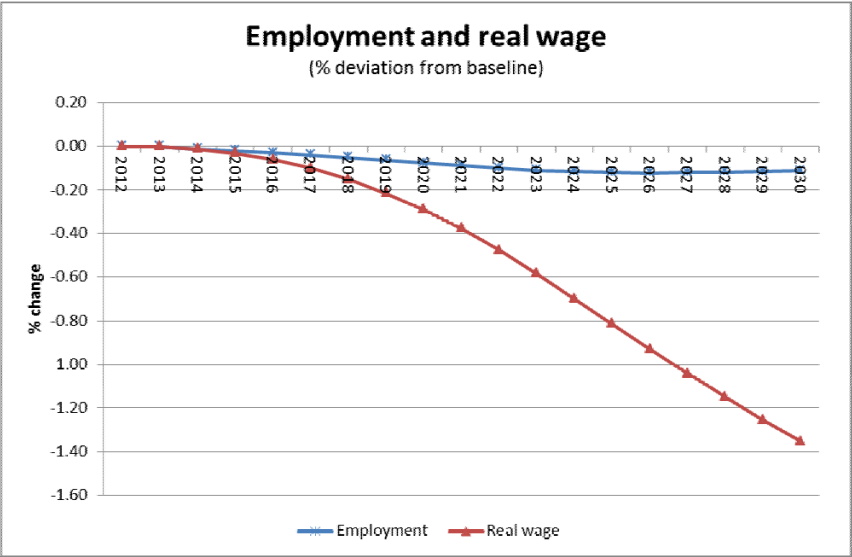
Navigate the directory to the folder called **EX45**. You do not have to rename the zip file. Click on **Save**. A box will appear with the following message: "Do you want to add some other extra files to the ZIP archive?...". Click on **No** and then **OK**. Your ZIP file is created and stored in the EX45 folder.

You may also want to save the simulation results. To do this click on **Zip | Zip up all results** or choose **Zip up selected results**. For this exercise pick Zip up all results. You have to navigate the directory to the folder called EX45 and pick a name for results zip.

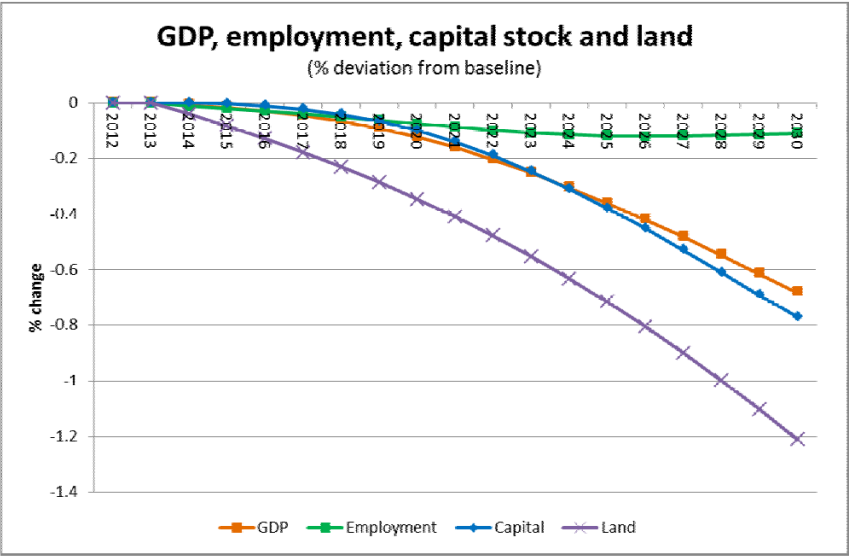


Be careful!. You will see that Rundynam automatically adds a **-res1.zip** in the file name. This is a way to show that this is a zip file containing results. Do not delete **-res1**. Save the file as EX45-res1.zip and click on save. It will take a few seconds to save all the simulation results. Finally a message will appear stating that the Results Zip file was created. Click on **OK**.

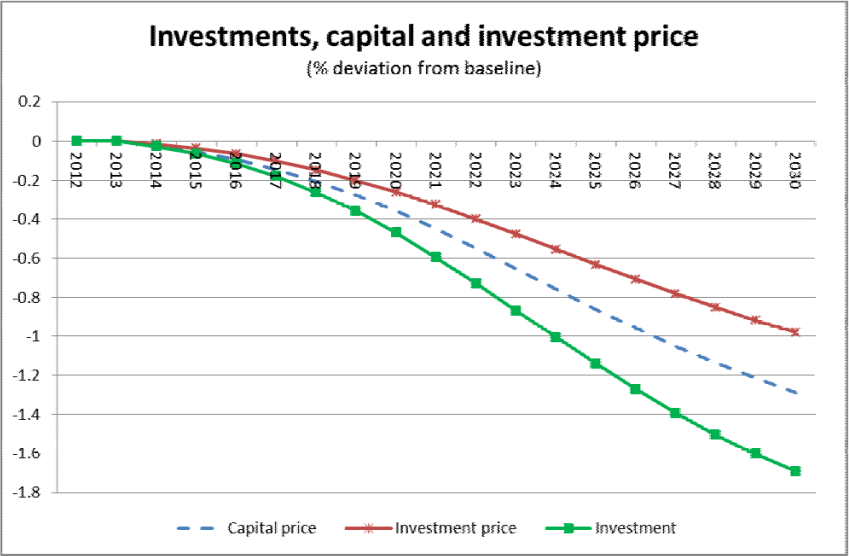
Graph 3.2.



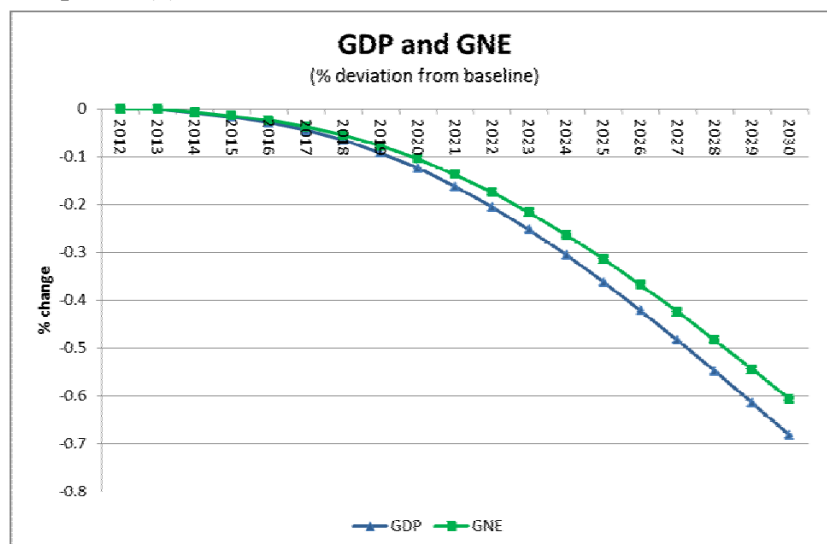
Graph 3.3.



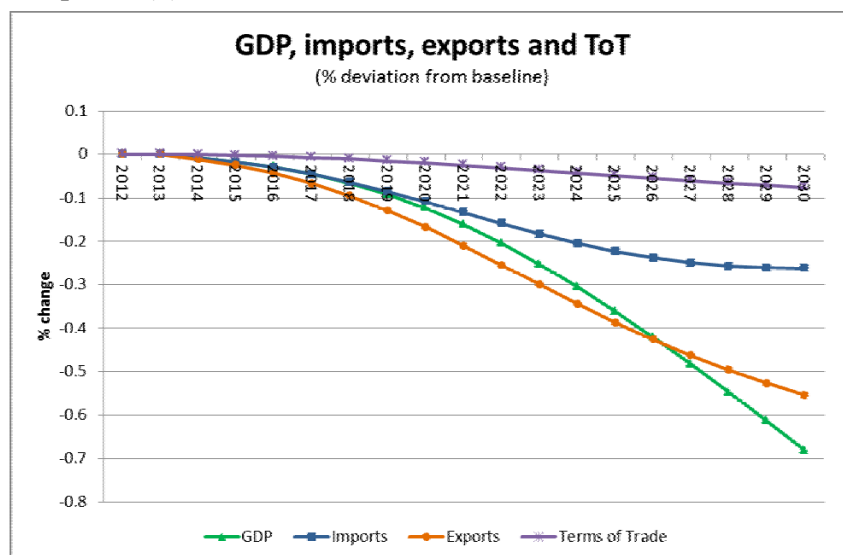
Graph 3.4



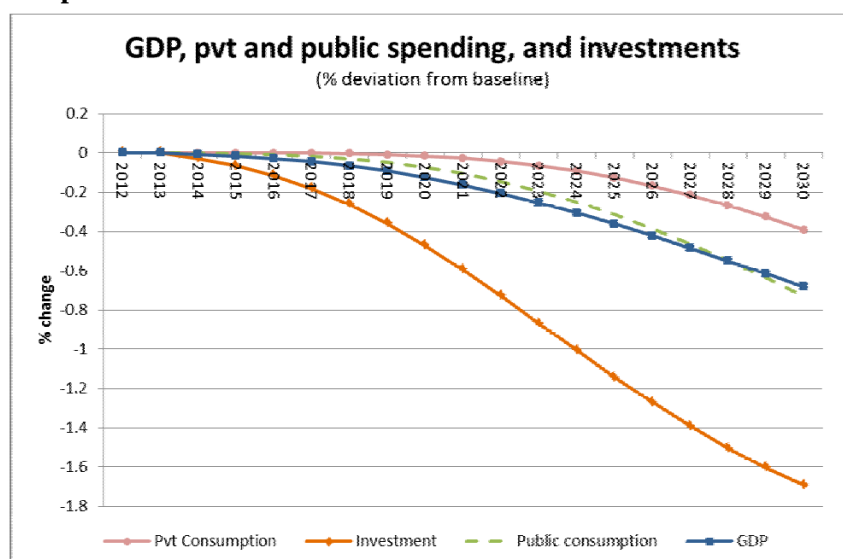
Graph 3.5 (a).



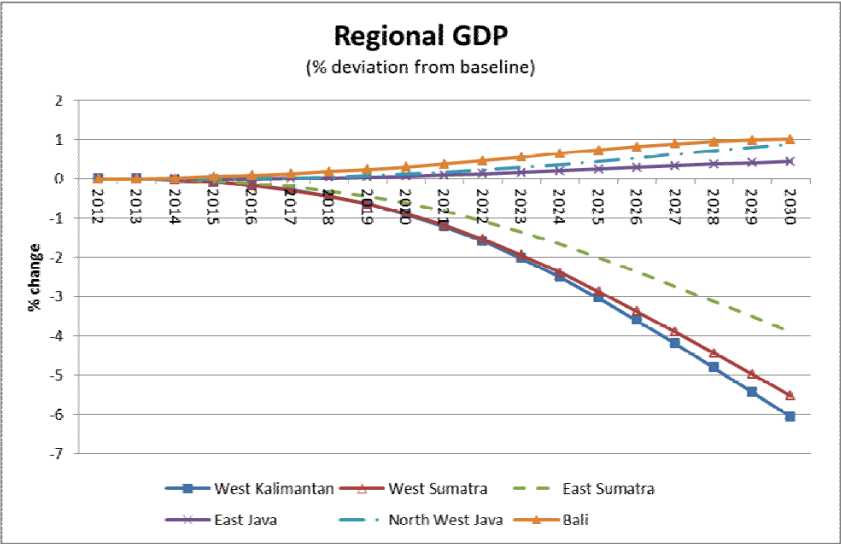
Graph 3.5 (b).



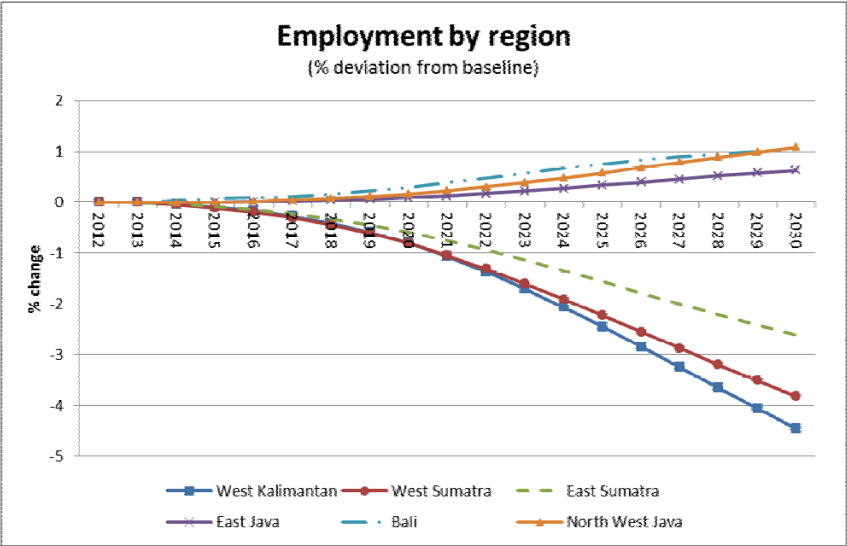
Graph 3.6.



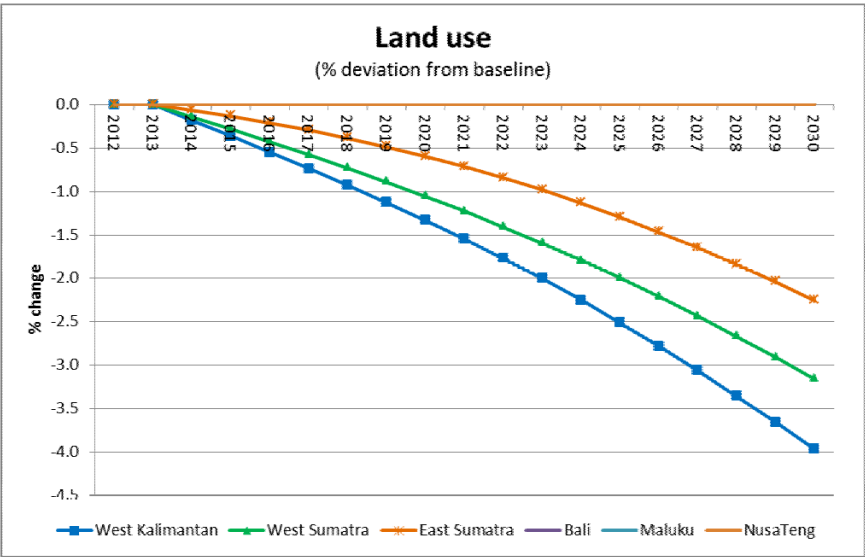
Graph 3.7.



Graph 3.8 (a).



Graph 3.8 (a).



Graph 3.9.

